

JMO – 2016

Answer all questions. All questions carry equal marks.

Full Marks 100

1.

(a) Determine which is greater between: 99^{100} or 100^{99} .

(b) Find the value of: $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} + \frac{1}{90} + \frac{1}{110} + \frac{1}{132} + \frac{1}{156} + \frac{1}{182} + \frac{1}{210}$.

2.

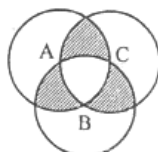
(a) Find the value of: $\sqrt{44444444444444444444 - 8888888888}$

(b) If a rebate of 4% is allowed on selling price of an article one received 8% profit. If no rebate is allowed on the sale price then what will be the percentage of profit?

3.

(a) Determine which of the fraction is greater: $\frac{4567890123}{780123456}$ or $\frac{4567890124}{780123458}$

(b) Three congruent circles each of radius 14 cm with centre A, B, C respectively are drawn such that each passes through the centre of the other two circles as shown in the figure. Find the area of the shaded portion.



4. The measure of the smallest angle formed between the hour hand and the minute hand of a clock at exactly 9 AM is 90° . Determine after exactly what time the acute angle between the two hands will be 90° .

5. When the first digit from left of a four digit number is shifted to the unit's place then the new number so obtained is larger than original number by 4707. Find the original number.

6.

(a) Find the remainder when 3^{2016} is divisible by 7.

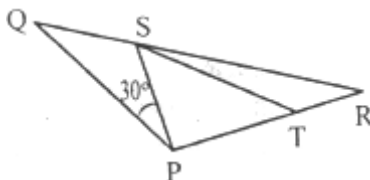
(b) Prove that for every positive integer n , $\frac{n+7}{2n+13}$ is a fraction in lowest term.

7. The cost of 3 apples, 4 bananas and 2 coconuts is Rs. 30 and that of 2 apples, 5 bananas and 1 coconut is Rs. 28, find the total cost of 4 apples, 17 bananas and 1 coconut.

8. N is the largest number such that $14N$ be a 100 digit number. What is the 64^{th} digit of N counted from right?

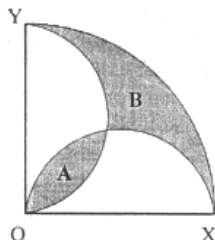
9. Two candles of same heights take 3 hrs and 4 hrs to finish burning respectively (They have different diameters). At what time must the candle be lit that the ratio of their height will be 1:2 at 4.00 PM?

10. In the figure triangle PQR has $PQ = PR$, $\angle QPS = 30^\circ$. If $PS = PT$, then determine $m\angle TSR$.



11.

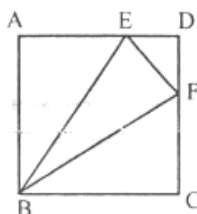
- (a) Mahesh sold 60% less number of soft drink bottles than mineral water bottles on Dusshera. The profit of each bottle of mineral water is 70% less than each bottle of soft drinks. The profit made from selling the soft drink was Rs. 1200. What was the profit made from the sale of mineral water?
- (b) In the given figure, OXY is the $\frac{1}{4}$ th of a circle representing one quadrant. Taking OX and OY as diameters semicircles are drawn. The area of shaded portions are written in the form A and B. The find the ratio of A and B.



12. In the below multiplication determine the letter represent which digit, where each letter represent a distinct digit.

$$\begin{array}{r} \text{A B C D E F} \\ \times \quad \quad \quad \text{F} \\ \hline \text{G G G G G G} \end{array}$$

13. ABCD is a square. E and F are points on AD and CD respectively. The area of triangle ABE is 5 square units. The area of triangle BCF is 4 square units. The area of triangle DEF is 3 square units. Find the area of the triangle BEF.

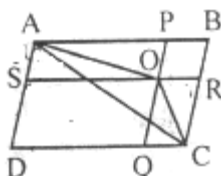


14. A cracker was found bursted in the staffroom during morning assembly of a school. Four probable culprits were called to the office by the teacher incharge of discipline for investigation.

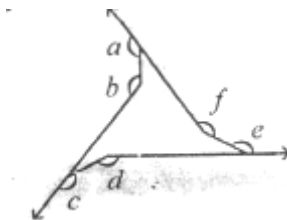
Amlan: Dayal bursted it
 Bhabani: Dayal bursted it.
 Chetan: I did not do it.
 Dayal: Bhabani is telling lies.

If only one suspected to have told the truth; then who was the culprit?

15. One day two brothers Ashok and Bikram were returning from school soon after the school closes by cycling with the speed of 140 m/min and 120 m/min respectively. At the same time their father cycled towards them at a speed of 160 m/min. He met Bikram 2 minutes after Ashok on the way and escorted them back to home. Determine the distance between the school and the home.
16. In the given diagram parallelogram ABCD has been divide into four parallelograms PQ and SR drawn parallel to BC and AB respectively and O is the point of intersection of PQ and SR. Prove that the area of SOQD – area of BPOR = 2 X area of triangle AOC.



17. In the given figure, the angle marked as a, b, c, d, e, f are congruent. Find the measure of each of them.



18. In triangle ABC , $\angle C$ is a right angle and CD is perpendicular to AB . Prove that:

$$\frac{1}{CD^2} = \frac{1}{AC^2} + \frac{1}{BC^2}.$$

19. In triangle ABC , $\angle ABC = \angle ACB = 78^\circ$, D and E are the points on AB and AC respectively such that $\angle BCD = 24^\circ$ and $\angle CBE = 51^\circ$, find $\angle BED$.
20. Solve for x, y, z :
- $$x + y + xy = 11$$
- $$y + z + yz = 19$$
- $$z + x + zx = 14$$
